THE UNIVERSITY OF SYDNEY

23rd December, 2025

RE: Submission of manuscript for publication in *Current Biology*

Dear Editors,

We wish to submit an original research article entitled **"Uncovering thousands of endosymbiont DNA transfer events within single cockroach genomes"** for your consideration. The study is authored by Kyle Ewart, Maxim Adams, Zhuzhi Zhang, Louise Baker, Kokuto Fujiwara, Yoshinobu Hayashi, Leo Featherstone, Oscar Lo Lu, Jil Helbling, Maya Moral, Kiyoto Maekawa, Harley Rose, Aaron Jex, Simon Ho and Nathan Lo. We believe this manuscript would be suitable for publication as a '*Report*.' This work has not been published elsewhere, nor is it currently under consideration for publication in any other journal.

Horizontal gene transfer (HGT) challenges conventional Darwinian evolutionary models by facilitating the movement of genetic material across species boundaries. Although HGT has been widely studied in eukaryotes, particularly in the context of bacterial endosymbionts, previous studies have generally identified relatively few transfers into host genomes (<170; with the exception of rotifers). Importantly, most of this work has focused on protein-coding genes; the extent and evolutionary significance of non-coding DNA transfer poorly understood.

In this study, we investigated HGT from the obligate bacterial endosymbiont *Blattabacterium cuenoti* into the genomes of its cockroach hosts. This symbiont is found in specialized cells of cockroach fat bodies and is inherited maternally though the eggs, providing the potential for recurrent transfer events. We analysed 14 cockroach and 4 termite genomes, including 8 newly generated long-read assemblies, and focused on putative HGT inserts ≥50 bp in both coding and non-coding regions. Using this approach, we identified tens of thousands of *Blattabacterium*-derived inserts across cockroach genomes (93–4900 per genome), orders of magnitude more than previously reported for any eukaryote outside rotifers. These inserts are typically short, and in some cases have persisted for more than 28 million years. We also identified numerous novel chimeric inserts composed of multiple segments originating from different regions of the *Blattabacterium* genome.

Together, our findings substantially revise current understanding of host–endosymbiont genome interactions by demonstrating pervasive horizontal transfer from a long-term obligate symbiont. The scale of transfer observed parallels that of mitochondrial DNA insertions (numts), highlighting similarities between contemporary endosymbiont–host interactions and the ancient bacterial endosymbiosis that gave rise to mitochondria. More broadly, this study provides a framework for uncovering previously hidden endosymbiont-derived DNA and advances our understanding of how horizontal transfer shapes eukaryotic genome evolution.

We would be pleased if you would consider our article for publication as a '*Report*' in *Current Biology*. We look forward to hearing from you.

Kind regards,

Kyle Ewart (on behalf of all co-authors)
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